

MEA를 사용한 in vitro 연구와 컴퓨터 연결

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Contents

1 The Neurally Controlled Animat

- MEA
- Neural-Computer Interface
- Controlling the Animat in its Virtual Environment
- Sensory Information and Feedback
- The Activity of the Animat

2 Biological Neural Cultures

- Free Energy Principle
- Dishbrain
- Network construction
- Result

The Neurally Controlled Animat

: Biological Brains Acting with Simulated Bodies

목표 및 특징:

- 배양하는 뇌로 가상 생물 (명명하여 Animat) 만들기
- 실제 생물 아냐 → sensory와 motor regions가 정해져 있지 아니함¹

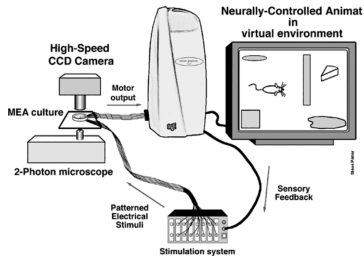


Figure 1: Scheme for the Neurally Controlled Animat. A network of hundreds or thousands of dissociated mammalian cortical cells (neurons and glia) are cultured on a transparent multi-electrode array. Their activity is recorded extracellularly to control the behavior of an artificial animal (the Animat) within a simulated environment. Sensory input to the Animat is translated into patterns of electrical stimuli sent back into the network.

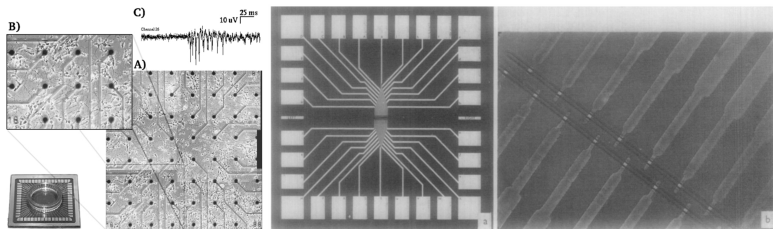
¹본 연구서 어떤 영향을 주는지 살펴보기

multi-electrode array (MEA)

- uses multi-electrode array (MEA) culture dishes
- to record neural activity from populations of neurons and use that activity to control an artificial animal
: 측정 & 입력
- 8×8 grid of 60- $10 \mu\text{m}$ -electrodes separated by $200 \mu\text{m}$
: $8 \times 8 - 4(\text{귀퉁이}) = 60$
- the 1.5-mm electrode region of the MEAs²
: $200 \times 7 + 10 \times 8 = 1480 \mu\text{m} \sim 1.5 \text{mm}$

²<https://www.sciencedirect.com/science/article/pii/S0165027001004125>

multi-electrode array (MEA)



(a) Paper

(b) Thomas et al., 1972

Figure 2: MEA 사진

- 뉴런들 투명 배양기 (clear substrate)서 자라 → MEAs allow detailed (submicron) optical investigation of neural connectivity (e.g., using 2-photon time-lapse microscopy) or submillisecond distributed activity (e.g., using voltage-sensitive dyes and a high-speed camera)

Neural-Computer Interface

- **구성**(참고: Figure 1):
 - **MEA culture**: animat의 뇌
 - **virtual environment**: animat의 서식지
 - **simulation system**: sensory 전달 시스템
- **animat 행동**: move forward, back, left or right³
- **뉴런 활동 측정**:
 - 증폭기와 A/D 변환기 측정
 - real-time data stream

2.95 MB per second (60 channels sampled at 25 KHZ)
 $: 2.95 \times 10^6 \times 8 / (60 \times 25000) = 15.733 \rightarrow$ 아마도 16bit로 샘플링?

³아마도 왼쪽, 오른쪽 회전? 아니 진짜 말 짜짜 모호해;;

Controlling the Animat in its Virtual Environment

A **clustering algorithm**⁴ was trained **to recognize spatio-temporal patterns** in the spike train, to use as **motor commands** for the Animat

$$A_n(t_i) = A_n(t_{i-1})e^{-\beta(t_i-t_{i-1})} + 1 \quad (1)$$

- i : each spike
- t_i : the time of the current spike
- n : the MEA channel number from 1 to 60
- β : a decay constant, $\beta = 1s^{-1}$
- vector A : the current spatial activity on the MEA

The activity vector A was then sent through a squashing function

$$P_n(t_i) = \tanh(\delta A_n(t_i)) \quad (2)$$

- δ : a gain factor, $\delta = 0.1$

⁴!!! 일반적인 clustering algorithm을 사용하지 아니한 이유를 고민해 보아요

Every 200 ms, the computer sampled the current pattern of normalized activity P (pattern)⁵

A pattern was grouped with the nearest cluster M_k if the Euclidean distance to that cluster was less than a threshold Δ , where $\Delta = 0.9$.

→ M_k 는 k 번째 cluster의 중심이겠구나!

$$M_k \Leftarrow \left(\frac{N_k}{N_k + 1} \right) + \left(\frac{P}{N_k + 1} \right) \quad (3)$$

$$N_k \Leftarrow N_k + 1 \quad (4)$$

- N_k : the number of occasions this pattern has matched
→ 점점 안정적인 M_k 값이 만들어 질 거야
- init value M_0 : the null cluster (all elements were set to zero) → No movement
- 각 cluster는 animat 행동을 임의로 지정함
- threshold값 보다 다 큼? → 해당 pattern으로 새로운 cluster 만들

⁵200ms마다? decay 요소인 β 를 포함한 e 의 지수 연산이 사용된 이유를 생각해 보아

Sensory Information and Feedback

cluster에 기반한 motor → virtual environment에서 animat을 조정 sensory

- 5개의 전극을 고르게 sensory 전극으로 설정
→ 충돌 전극, 앞, 뒤, 좌, 우 전극
- delivered four ± 400 mV, 200 μ s pulses

The Activity of the Animat

- 처음 8분: 피드백 없이 (sensory 없이) 그냥 클러스터링 진행
- 8분 이후: sensory 자극으로 피드백 (50분 진행, 비단 더 할 수 있음)

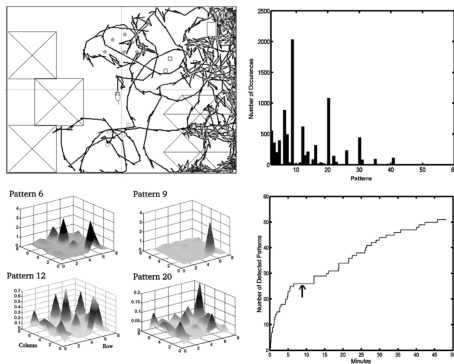


Figure 3: Movement and results of the session in which the Animat was online. Top left: Trajectory of movement of the Animat in a simulated room. Top right: The frequency that different patterns would recur. Bottom left: Examples of the four most common patterns. Bottom right: Number of novel patterns that the clustering algorithm detected as the session progressed (total 51). Bottom left: Examples of the four most common patterns.

???:

- 왜 map서 장애물을 넘어서 갈 수 있지?⁶
- 200ms마다 패턴 측정 → 클러스터링: 즉, 1초에 최대 5개의 새로운 클러스터가 만들어 질 수 있음
→ 비단! 13초를 봐

패턴은 가끔씩 2~3개의 순서로 발생하며, 몇 초 동안 반복 후 새로운 패턴이 만들어짐

→ 저자는 자연스럽게 또는 피드백에 따라 신경망이 진화하는 것이라고 해석

⁶논문서 object는 4개라고 한 적이 있었었었음

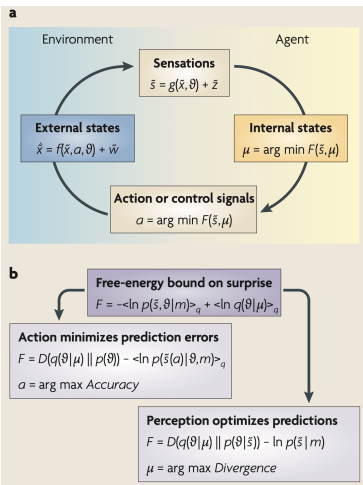
Dynamic Network Plasticity and Sample Efficiency in Biological Neural Cultures

: A Comparative Study with Deep Reinforcement Learning

일러두기

- 판단컨대, 논문 내용 짜짜해 :(근거가 부실부실합니다.
- 논문서 제시하는 근거를 비판적으로 바라보면 잼있을 것 같아요.
- Published 4 August 2025 (!!!)

Free Energy Principle



part a:

- $\mu(t)$: 뇌의 (내부) 상태
- $\tilde{s}(t)$: sensory signals: $\tilde{s}(t) = [s, s', s'', \dots]^T$
→ s' 은 이전 감각보다는 미분값인 것 같음(추정)
- $a(t)$: action
- $\vartheta \subset \{\tilde{x}, \theta, \gamma\}$: causes
- $\tilde{x}(t)$: hidden states, θ : parameters
- $\tilde{w}, \tilde{z}$: random fluctuations, γ : amplitude of $\tilde{w}, \tilde{z}$
- 윗첨자 *vector*, $\cdot$ 은 미분을 표현하는 것 같음(추정)
- F : free energy → surprise의 상한으로 생각해 보아

Figure 4 credit: <https://pubmed.ncbi.nlm.nih.gov/20068583/>

Figure 4: Free Energy Framework

Free Energy Principle

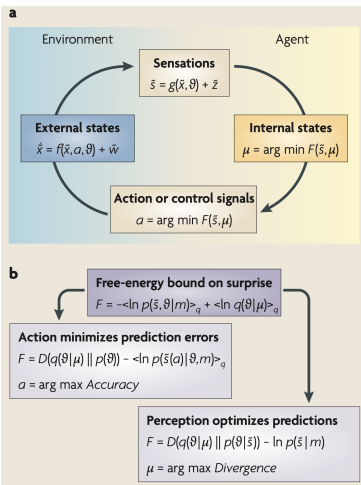


Figure 5: Free Energy Framework

part b:

- $q(\vartheta | \mu)$: recognition density: 뇌의 상태 $\rightarrow$ 원인 (!!!)
- $p(\tilde{s}, \vartheta | m)$: generates sensory samples and their causes, m : generative model (???)
- surprise(self-information): $I(x) = \log 1/P(x) = -\log P(x)$
- entropy: $\mathbb{E}[I(x)]$
- surprise: $-\ln p(\tilde{s}, \vartheta | m) = -\ln \int p(\tilde{s}, \vartheta | m) d\vartheta \rightarrow$ 원지 모를 복잡한 계산 $\rightarrow$ 상한 F
- KL divergence: divergence(차이), entropy의 비교: $D(P || Q)$
 F 를 KL divergence를 사용하여 전개
- Action minimizes prediction errors
 - KL divergence 부분: recognition density(prior의 표현을 생각해 보아!) an approximate posterior probability의 차이
 - $-\langle \ln p(\tilde{s}(a) | \vartheta, m) \rangle_q$: cross entropy! F 가 $\arg \min$ 연산에 따라 action은 cross entropy 값만을 바꿀 수 있음
현재 믿음 vs 이상적 사후확률 $\rightarrow$ cross entropy 값이 감소하며 실제 상한 값과 점점 가까워지고 Accuracy 증가해
- Perception optimizes predictions
 - $-\ln p(\tilde{s} | m)$: surprise, perception으로 인한 F 최소화에선 μ 가 없어서 변화하지 아니해
 - KL divergence 부분: 최대화 되어 $\rightarrow$ divergence(차이)가 최대화 되어
현재 믿음 vs 사전 믿음
 $\rightarrow$ 차이가 큰 것은 구별이 용이한 것, 즉 불확실하고 예측 불가능한 것보다 확실한 믿음을 만들 수 있는 예측 가능한 것을 의미함

Dishbrane

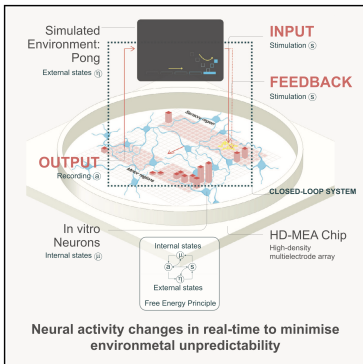


Figure 6: Dishbrain architecture^a

^aKagan, Brett J. et al. 2022

- free energy principle에 기반하여 in vitro로 dishbrain을 학습시켜 pong 게임을 성공적으로 학습함 (시청각 자료 여기 클릭)
- free energy principle 요소:
 - μ : in vitro neurons
 - a : action: from motor regions
 - x : simulated environment stats: pong game
 - s : sensory: to sensory regions
 - 학습 요소: sensory에 predictable or unpredictable feedback 전달 → 게임 결과에 따른 divergence 조절

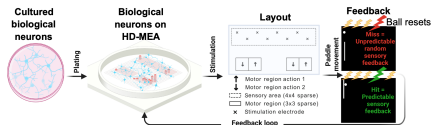
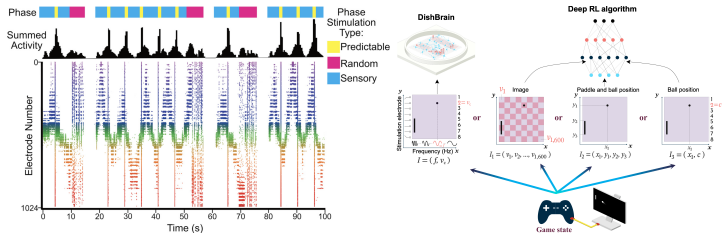


Figure 7: Schematic illustration of the DishBrain feedback loop

- Predictable feedback: 150 mV and a frequency of 5 Hz
- Unpredictable feedback: 4초 동안 8개 모두에



(a) 1024개의 전극⁷의 스파이크 여부⁸ (b) dishbrain과 RL 기법에서의 정보 입력 방법

Figure 8: Dishbrain을 이해하기 좋은 figure들

좌측 그림서 공을 정확히 바로 맞춘 경우(hit)와 그렇지 아니한 경우(miss)의 스파이크를 비교하면 입력의 예측 가능성의 의미를 알 수 있음
우짜 그림서 Dishbrain 부분을 보면 virtual environment(즉, pong gmae)의 상태가 어떻게 dishbrin으로 전달 되는지 알 수 있음

⁷HD-MEA의 전극 수

⁸시청각 자료 보면 끝 부분 맞는 것도 약간의 불확실성을 주어야 할 듯..

Network construction

목표: 스파이크들을 가지고 네트워크를 추정

→ 해당 네트워크를 rest 상태 및 RL 모델들의 네트워크와 비교 (???)

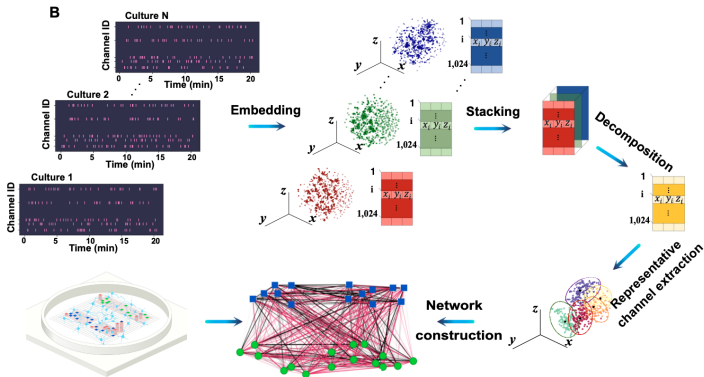


Figure 9: A schematic illustration of the overall network construction framework

- embedding: t-distributed stochastic neighbor embedding (t-SNE): 3d 로 변환 (차원 축소)
- 여러 culture서 나온 데이터를 embedding한 것을 합쳐서 tensor로
- Tucker decomposition로 행렬 분해
- K-medoids algorithm으로 30개 군집화
- Pearson correlation로 node간 weight 결정

Result

Gameplay vs Rest

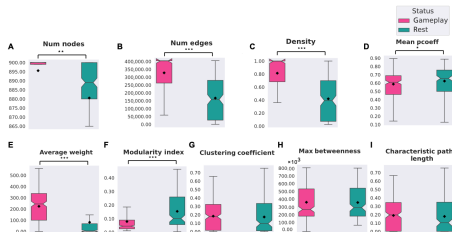


Fig. 2. Significant network plasticity occurs in biological cultures when embodied in the game environment. (A to I) Network summary statistics of 1,024 recorded channels using the full duration of all Gameplay and Rest sessions. Using one-way t tests, we found significant differences in the number of nodes ($P = 3.072e-23$), number of edges ($P = 8.396e-26$), density ($P = 1.009e-25$), mean participation coefficient (pcoeff) ($P = 3.400e-2$), average weight ($P = 8.910e-20$), and modularity index ($P = 4.129e-13$) between Gameplay and Rest. No significant differences were found for clustering coefficient ($P = 0.968$), max betweenness ($P = 0.890$), or characteristic path length ($P = 0.533$).

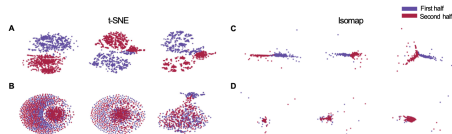


Fig. 3. Low-dimensional representation of 3 samples of (A) Gameplay sessions and their following (B) Rest sessions using t-SNE as well as (C) Gameplay sessions and their following (D) Rest sessions using Isomap. The purple and maroon dots are the channel representations in the embedding space in the first and second halves of the recordings, respectively. Both dimensionality reduction algorithms were able to distinguish between the 2 halves of recording during Gameplay but not during Rest sessions.

Figure 10: Result 1

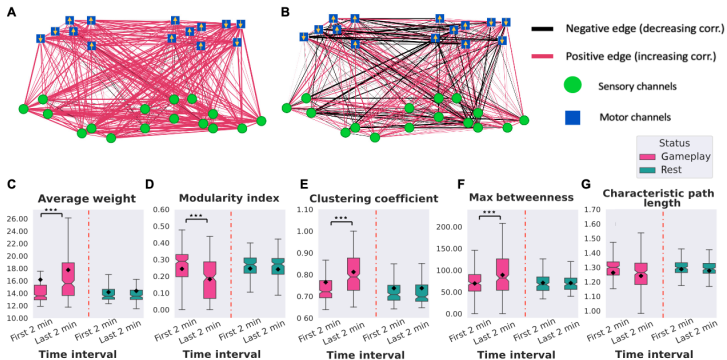


Fig. 4. The average connectivity networks using 30 representative channels over all (A) Gameplay and (B) Rest sessions with edge weights representing changes in functional connectivity between channel pairs when comparing the last 2 min to the first 2 min of recordings. Edge colors signify the direction of these connectivity changes, with red indicating increases and black indicating decreases. Motor and sensory region channels are represented by blue squares and green circles, respectively. Arrows on motor region nodes show the paddle's movement direction as per their position in the predefined layout in Fig. 1. (C to G) Network summary statistics between the first and last 2 min of Gameplay and Rest recordings using the 30 representative channels in the lower-dimensional space. All of these metrics except the characteristic path length showed statistically significant differences using one-way ANOVA during Gameplay ($P = 2.265e-3$, $P = 8.478e-8$, $P = 1.891e-6$, $P = 1.005e-4$, and $P = 0.071$, respectively), but not in the Rest condition of the cultures ($P = 0.864$, $P = 0.670$, $P = 0.738$, $P = 0.281$, and $P = 0.899$, respectively). *** $P < 0.001$.

Figure 11: Result 2

Dishbrain vs RL

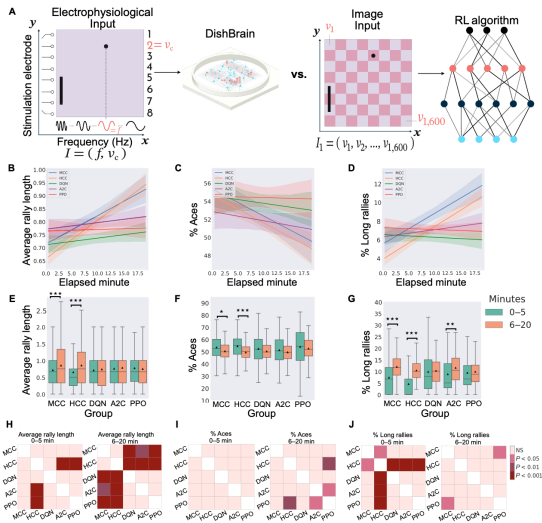


Figure 12: Result 3

나의 생각

해당 figure 말고도 관련 result와 관련된 자료는 많이 있음. 비단!

- vs Rest: rest 상태는 sensory regions에 전압을 주지 아니함. 차이가 있는 것이 당연함.
- vs RL: 차원 축소 등 특징적인 요소만을 가져오고 변환해 만든 네트워크. 뇌가 가지는 인공지능과 다른 요소를 반영하지 아니하며, 실제로는 800,000개의 뉴런이 dishbrain에 있다고 함. 즉 RL과 node, edge, weight을 비교하는 것은 아무런 의미가 없음.